

2-14 Fluently Add Multi-Digit Numbers

Learning goal: I can add multi-digit numbers by decomposing an addend, using partial sums, or using an algorithm with regrouping.

Textbook: pages 87–90. The book adds $2,322 + 1,569$ by decomposing an addend, with partial sums, and with an algorithm.

Words to Know

decompose an addend	Break one addend into thousands, hundreds, tens, and ones and add them one at a time.
partial sums	Add each place separately, then add those answers.
algorithm	Step-by-step column adding: start with the ones and move left.
regroup	Trade 10 of one place for 1 of the next place: 15 ones = 1 ten and 5 ones.

How to Teach It

- 1. Decompose an addend:** keep the first number. Add the thousands, then hundreds, tens, and ones of the second number, keeping a running total.
- 2. Partial sums:** add each place and write each answer with its full value ($300 + 500 = 800$, not 8). Then add the partial sums.
- 3. Algorithm:** line up the numbers by the **ones**. Add the ones first. If a place adds to 10 or more, write the ones digit and regroup 1 to the next place (write a small 1 above it).
- If the last place adds to 10 or more, the regrouped 1 makes a new place ($5,736 + 4,582 = 10,318$).

Tip: Any strategy is fine. Ask, “What did you do with the extra ten?” In stories, cross out numbers you do not need; for “fewer than,” decide who has more.

Worked Examples

Example 1: Decompose an addend

$3,146 + 2,537$: add 2,000, then 500, then 30, then 7.

$3,146 \rightarrow 5,146 \rightarrow 5,646 \rightarrow 5,676 \rightarrow \mathbf{5,683}$

Example 2: Partial sums

$$\begin{array}{r}
 2,457 \\
 + 1,368 \\
 \hline
 2,000 + 1,000 = 3,000 \\
 400 + 300 = 700 \\
 50 + 60 = 110 \\
 7 + 8 = 15 \\
 \hline
 3,825
 \end{array}$$

Example 3: Use an algorithm

$ \begin{array}{r} 11 \\ 2,457 \\ + 1,368 \\ \hline 3,825 \end{array} $	Ones: $7 + 8 = 15$. Write 5, regroup 1 ten. Tens: $1 + 5 + 6 = 12$. Write 2, regroup 1. Hundreds: $1 + 4 + 3 = 8$. Thousands: $2 + 1 = 3$.
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Example 4: A new place; a “fewer” story

$5,736 + 4,582$: thousands $1 + 5 + 4 = 10$, so the sum is **10,318**.

Ben has 1,250 fewer cards than Kai. Ben has 2,430. Kai has more, so add: $2,430 + 1,250 = \mathbf{3,680}$.

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|--|--|
| ☐ Decompose an addend and keep a running total. | ☐ Line up numbers with different numbers of digits. |
| ☐ Find partial sums with full place values. | ☐ Judge whether someone’s strategy works. |
| ☐ Use the algorithm and regroup correctly. | ☐ Solve “fewer than,” extra-information, and two-step problems. |

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Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Use the algorithm. Write the regrouped 1s above the numbers.

$$\begin{array}{r} 548 \\ + 376 \\ \hline \end{array}$$

- B.** Mia and Jon find $4,256 + 5,368$. Mia uses an algorithm. Jon uses partial sums. Can either strategy be used? Explain. Find the sum.

Sum: _____

- C.** Add $2,316 + 587$ by decomposing 587. Hint: what can you add to 2,316 to make a hundred?

Sum: _____

On My Own *Try these by yourself. Show your work.*

- 1.** Solve. Use any strategy.

$$467 + 385 = \underline{\hspace{2cm}}$$

- 2.** Solve. Use any strategy.

$$\underline{\hspace{2cm}} = 125 + 4,631$$

- 3.** Solve. Use any strategy.

$$\underline{\hspace{2cm}} = 4,968 + 3,275$$

- 4.** Solve. Use any strategy.

$$31,402 + 5,375 = \underline{\hspace{2cm}}$$

5. Nia adds $2,639 + 457$. She adds 61, then 400, then 6. Do you agree with her strategy? Explain. What is the correct sum?

Sum: _____

6. Sam adds $548 + 263$. His partial sums are shown. What is another way Sam could find the sum? Show it.

$$500 + 200 = 700$$

$$40 + 60 = 100$$

$$8 + 3 = 11$$

7. Leo walked 2,165 fewer steps than his sister Ana. Leo walked 2,748 steps. How many steps did Ana walk?

Steps: _____

8. A library lent 1,245 books in fall and 2,136 books in winter. In spring it lent 1,105 more books than in fall and winter together. How many books did it lend in spring?

Books: _____

9. Solve. Explain your strategy.

$$538 + 274 = \underline{\hspace{2cm}}$$

Strategy: _____

10. Solve. Explain your strategy.

$$6,247 + 568 = \underline{\hspace{2cm}}$$

Strategy: _____

11. Mr. Diaz needs 800 paper cups. Cups come in packs of 300. How many packs should he buy?

Packs: _____

12. A school raised \$3,276 at a fair and \$1,858 at a bake sale. It spent \$900 on books. How much money did the school raise?

Raised: _____

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Together

A	924 <i>Ones $8 + 6 = 14$ (write 4, regroup 1). Tens $1 + 4 + 7 = 12$ (write 2, regroup 1). Hundreds $1 + 5 + 3 = 9$. Common mistakes: 814 (forgetting both regrouped 1s) or 914 / 824 (forgetting one).</i>
B	Yes, both work. Sum: 9,624. <i>Algorithm: $14 \rightarrow 4$ r1; $1 + 5 + 6 = 12 \rightarrow 2$ r1; $1 + 2 + 3 = 6$; $4 + 5 = 9$. Partial sums: $9,000 + 500 + 110 + 14 = 9,624$.</i>
C	$2,316 + 84 = 2,400$; $2,400 + 500 = 2,900$; $2,900 + 3 = \mathbf{2,903}$ <i>$84 + 500 + 3 = 587$, so the parts add back up. Place-value parts (500, 80, 7) also work.</i>

On My Own

1	852 <i>Items 1–4: any strategy. $7 + 5 = 12$ r1; $1 + 6 + 8 = 15$ r1; $1 + 4 + 3 = 8$.</i>
2	4,756 <i>Line up the ones: 125 goes under the 631 part.</i>
3	8,243 <i>Regroups three times: 13, 14, 12.</i>
4	36,777
5	No. $61 + 400 + 6 = 467$, not 457, so she adds 10 too many (she would get 3,106). Correct sum: $2,639 + 457 = \mathbf{3,096}$. <i>Her idea of making a friendly number works; the parts just have to add up to 457 (for example, $61 + 396$).</i>
6	Sample: the algorithm. $8 + 3 = 11$ (write 1, regroup 1); $1 + 4 + 6 = 11$ (write 1, regroup 1); $1 + 5 + 2 = 8$. Sum 811 . <i>Also accept decomposing an addend: $548 + 200 = 748$, $+ 60 = 808$, $+ 3 = 811$.</i>
7	Ana walked more. $2,748 + 2,165 = \mathbf{4,913}$ steps. <i>Subtracting is a common mistake with “fewer than.”</i>
8	$1,245 + 2,136 = 3,381$. $3,381 + 1,105 = \mathbf{4,486}$ books. <i>Two steps: add fall and winter first, then add 1,105 more.</i>
9	812 <i>e.g., partial sums $700 + 100 + 12 = 812$.</i>
10	6,815 <i>e.g., algorithm: 15 r1, 11 r1, $1 + 2 + 5 = 8$, 6.</i>
11	$300 + 300 = 600$ is not enough; $600 + 300 = 900$ is enough. 3 packs.
12	$\$3,276 + \$1,858 = \mathbf{\$5,134}$ <i>The \$900 is extra information.</i>